

EXPERIMENT-15

Implement Iris Flower Classification using Naive Bayes.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

print("=" * 70)
print("    IRIS CLASSIFICATION USING NAIVE BAYES")
print("=" * 70)

uploaded = files.upload()

filename = list(uploaded.keys())[0]
data = pd.read_csv(filename)

print("\nDataset:")
print(data.head(10))

X = data.iloc[:, 1:5]
y = data.iloc[:, 5]

encoder = LabelEncoder()
y = encoder.fit_transform(y)
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=42
)

model = GaussianNB()
print("\nTraining Naive Bayes Classifier...")
model.fit(X_train, y_train)
prediction = model.predict(X_test)
print("\nTraining Completed.")
print("\nActual Values:")
print(list(y_test))
print("\nPredicted Values:")
print(list(prediction))
print("\nAccuracy:")
print(round(accuracy_score(y_test, prediction) * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, prediction))
sample = [[6.1, 2.8, 4.7, 1.2]]
result = model.predict(sample)
print("\nNew Flower:")
print(sample)
print("\nPredicted Flower:")
print(encoder.inverse_transform(result)[0])

```

Sample Input:

```

... =====
                IRIS CLASSIFICATION USING NAIVE BAYES
                =====
Choose Files Iris.csv
Iris.csv(text/csv) - 5107 bytes, last modified: 8/5/2026 - 100% done
Saving Iris.csv to Iris (1).csv

```

Sample Output:

```
... Dataset:
  Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0  1      5.1         3.5         1.4         0.2  Iris-setosa
1  2      4.9         3.0         1.4         0.2  Iris-setosa
2  3      4.7         3.2         1.3         0.2  Iris-setosa
3  4      4.6         3.1         1.5         0.2  Iris-setosa
4  5      5.0         3.6         1.4         0.2  Iris-setosa
5  6      5.4         3.9         1.7         0.4  Iris-setosa
6  7      4.6         3.4         1.4         0.3  Iris-setosa
7  8      5.0         3.4         1.5         0.2  Iris-setosa
8  9      4.4         2.9         1.4         0.2  Iris-setosa
9 10      4.9         3.1         1.5         0.1  Iris-setosa

Training Naive Bayes Classifier...

Training Completed.

Actual Values:
[np.int64(1), np.int64(0), np.int64(2), np.int64(1), np.int64(1), np.int64(0), np.int64(1), np.int64(2), np.int64(1), np.int64(1), np.int64(2), np.int64(0), np.int64(0), np.i

Predicted Values:
[np.int64(1), np.int64(0), np.int64(2), np.int64(1), np.int64(1), np.int64(0), np.int64(1), np.int64(2), np.int64(1), np.int64(1), np.int64(2), np.int64(0), np.int64(0), np.i

Accuracy:
97.78 %

Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

New Flower:
[[6.1, 2.8, 4.7, 1.2]]

Predicted Flower:
Iris-versicolor
```